Regular Expressions in Automata Theory

Regular expressions are symbolic notations used to define search patterns in strings. They describe regular languages and are commonly used in tasks such as validation, searching, and parsing.

The symbols are identical which are used to construct the languages, and any given expression that has a language closely associated with it.

A **Regular Expression** can be recursively defined as follows −

- **ε** is a Regular Expression indicates the language containing an empty string. **(L (ε) = {ε})**
- **ϕ** is a Regular Expression denoting an empty language. **(L (ϕ) = { })**
- **x** is a Regular Expression where **L = {x}**
- If **X** is a Regular Expression denoting the language **L(X)** and **Y** is a Regular Expression denoting the language **L(Y)**, then
    - **X + Y** is a Regular Expression corresponding to the language **L(X) ∪ L(Y)** where **L(X+Y) = L(X) ∪ L(Y)**.
    - **X . Y** is a Regular Expression corresponding to the language **L(X) . L(Y)** where **L(X.Y) = L(X) . L(Y)**
    - **R*** is a Regular Expression corresponding to the language **L(R*)** where **L(R*) = (L(R))***
- If we apply any of the rules several times from 1 to 5, they are Regular Expressions.

Properties of Regular Expressions in TOC

All the properties held for any regular expressions R, E, F and can be verified by using properties of languages and sets.

Additive (+) Properties

The additive properties of regular expressions are as follows −

R + E = E + R

R + Ø = Ø + R = R

R + R = R

(R + E) + F = R + (E + F)

Product (.) Properties

The product properties of regular expressions are as follows −

RØ = ØR = Ø

RΛ = ΛR = R

(RE)F = R(EF)

Distributive Properties

The distributive properties of regular expressions are as follows −

$R(E + F) = RE + RF$

$(R + E)F = RF + EF$

Closure Properties

The closure properties of regular expressions are as follows −

$\emptyset^* = \wedge^* = \wedge$

$R^* = R^*R^* = (R^*)^* = R + R^*$

$R^* = \wedge + RR^* = (\wedge + R)R^*$

$RR^* = R^*R$

$R(ER)^* = (RE)^*R$

$(R + E)^* = (R^*E^*)^* = (R^* + E^*)^* = R^*(ER^*)^*$

All the properties can be verified by using the properties of languages and sets.